COMP 364/464: High Performance Computing
Project 1: HPC Systems Survey
Term: Fall 2026
Instructor: Melanie Cornelius

Assigned: 9 September 2026
Due: 26 September 2026, 11:59 PM AoE

The purpose of this assignment is two-fold: first, to get familiar with more HPC systems, past and present, and second, to practice survey writing.

Survey 5 to 10 machines from the Top500 list (past or present), find papers or articles on them, describe the machines, and discuss their properties, trends, and use-cases. Pay attention to anything interesting about the machine. Did it hit the top 10 in the Top500? Did it use a new interconnect, memory technology, or GPU? Why did that machine matter at the time?

In addition, you should **select a subset of machines with intent.** What do you find interesting about large-scale computing? Look into machines that fit your interests: for example, GPU-heavy machines, ARM machines, machines intended to do a specific type of science, machines with novel interconnects, etc. I'd like you to do this assignment with a purpose in mind, not just randomly selecting ten machines from a list. If you're stuck, reach out on Piazza, or read into some of the machines we've covered in class for ideas!

Format your paper however you like, but **make sure it's legible and logical**. Use subsectioning and images when appropriate. I recommend using one of the standard paper formats, such as the IEEE manuscript templates, but you are not restricted to a formal paper format.

Make sure you **use reliable sources** wherever possible; papers (whitepapers or peer-reviewed) are ideal, and articles from labs or facilities are good standards. Some machines are old, so I understand published works aren't always possible, but **you must at minimum correctly cite your sources**, and you should at least try to find formal publications.

An example survey will be posted with a graded rubric. An example of responsible AI use reporting will also be included.

For 5% extra credit, use LaTeX and Overleaf!

Different sections:

- Students in the 364 section should select at least 5 systems and write at least 2,000 words.
- Students in the 464 section should select at least 7 systems and write at least 3,000 words.

Topic	Points	Full credit	Partial	Minimal
Scope & selection	10	Minimum number of machines chosen (COMP 364: 5+ systems, COMP 464: 7+ systems) with a purpose or rationale; the subset is coherent (spans vendors, architectures, or eras, or is a deliberate themed slice).	Right count but the rationale is thin, or the subset feels arbitrary.	Too few machines, or no reasoning given for the choice.
Technical description: architecture & interconnect	30	Each machine's compute (CPU/GPU/accelerator, node design, memory) and its network (topology and technology, e.g. Slingshot, InfiniBand, custom dragonfly or torus) described correctly, with terms used accurately.	Mostly correct but shallow.	Errors in the facts, or the architecture or network is largely ignored.
Sources	15	Credible, relevant sources for each machine (architecture papers, vendor or system docs, SC/ISC papers, Top500 data), not just marketing pages.	Sources present but uneven, or over-reliant on a single type.	Few, weak, or missing sources.
Synthesis & discussion	30	Goes beyond machine-by-machine: compares across the set, identifies trends, trade-offs, and gaps.	Some comparison, but the writing stays mostly descriptive.	A list of machines with no cross-cutting analysis.
Organization & clarity	10	Clear scope statement, logical structure, plain and readable prose. Length reached (COMP 364: 2k words, COMP 464: 3k words)	Structure loose, or clarity is low. Length not reached.	Hard to follow. Length not reached.
Citations	5	Consistent, correct attribution; every source traceable.	Inconsistent or incomplete attribution.	Missing or improper citations.
Total	**100**			